

Josh Chung — Mock Interview Prep

Role: Engineer, Field Service (Early Career) | Candidate: BS Mechanical Engineering, UC Riverside (June 2025)

1. Ten Behavioral Questions (Mapped to Competencies)

Problem-Solving & Technical Judgment

1. "Tell me about a time a prototype or test didn't perform as expected. Walk me through how you diagnosed the problem and what you did next."

Problem-Solving & Technical Judgment

2. "Describe a situation where you had to follow strict safety or compliance protocols during a hands-on project. How did you ensure nothing was overlooked?"

Communication & Influence

3. "Give me an example of when you had to explain a technical decision or trade-off to someone outside your immediate team — a professor, sponsor, or external reviewer. How did you make it clear?"

Drive & Motivation

4. "Tell me about a project where you went through multiple design or test iterations. What drove the changes between each version?"

Communication & Influence

5. "Describe your experience creating or interpreting technical documentation — drawings, BOMs, assembly procedures. How did you ensure accuracy?"

Drive & Motivation

6. "Walk me through a time you personally fabricated, assembled, or built something. What tools or processes did you use, and what challenges came up during the build?"

Teamwork & Collaboration

7. "Tell me about a time you had to coordinate with teammates to meet a hard deadline on a technical project. What was your role and how did you handle disagreements or roadblocks?"

Problem-Solving & Technical Judgment

8. "You have a Green Belt certification. Give me a specific example of when you applied a structured problem-solving or continuous improvement method to a real project."

Adaptability & Learning

9. "Describe a time you had to quickly learn a new tool, technology, or process you hadn't used before. How did you get up to speed?"

10. "Tell me about a situation where you had to decide whether to solve a problem on your own or escalate it to someone more experienced. What factors drove your decision?"

Uncovered competencies: Leadership & Initiative, Failure & Self-Awareness — neither maps cleanly to the original 10 without forcing it.

2. Key Qualities the Role Demands

- Troubleshooting and diagnostic thinking
- Safety discipline
- Hands-on willingness to get sleeves rolled up
- Customer communication and service mindset
- Knowing when to escalate vs. solve independently
- Working independently without constant oversight
- Adaptability to unfamiliar environments (cleanrooms, customer sites, travel)

3. Josh's Strongest Resume Experiences for This Role

- **NASA prototype iterations (3+ iterations, 3D printing, CNC machining)** — directly shows hands-on fabrication and iterative troubleshooting
- **NASA EVA glove compatibility / interface constraints** — designing to strict external specs mirrors working within customer fab requirements
- **NASA FEA testing and design verification in low-gravity specs** — safety-critical validation against strict specifications
- **Bridge incremental load testing and failure analysis** — diagnosing where and why something breaks, core field service skill
- **Bridge 3+ iterations refining truss geometry** — continuous improvement loop, maps to the CI mention in the posting
- **Robotics gripper — learning Roboflow/computer vision tools and refining through additive manufacturing** — adaptability to unfamiliar tech and hands-on iteration combined
- **GD&T drawings, assembly procedures, and BOM creation** — technical documentation is explicitly called out in "what sets you apart"
- **NASA decision matrices and trade-off documentation** — communicating engineering rationale to stakeholders, maps to customer-facing explanation skills

4. Deep Dive — Questions 2, 3, and 10

These are the highest-priority questions for the field service role. Safety, communication, and escalation judgment are the core differentiators.

Question 2 — Safety / Compliance Protocols

"Describe a situation where you had to follow strict safety or compliance protocols during a hands-on project. How did you ensure nothing was overlooked?"

They're looking for:

- Specific awareness of *why* the protocol exists, not just blind compliance
- A system or habit for ensuring nothing gets skipped (checklists, peer review, documentation)
- Evidence that safety isn't something imposed on you — it's something you own

They don't want to see:

- Vague answers like "I always follow the rules"
- No concrete example or process described
- Treating safety as an inconvenience or afterthought
- Taking credit for safety culture without describing personal action

Best Josh experiences:

- NASA design verification against strict specs for low-gravity operation — safety-critical testing with clear external standards
- NASA interface constraints and EVA glove compatibility — designing within non-negotiable safety/usability boundaries

Question 3 — Explaining Technical Decisions to External Audiences

"Give me an example of when you had to explain a technical decision or trade-off to someone outside your immediate team — a professor, sponsor, or external reviewer. How did you make it clear?"

They're looking for:

- Ability to translate engineering rationale into terms the audience actually cares about
- Awareness of *who* the audience is and adjusting accordingly
- Structured communication — not rambling through every detail

They don't want to see:

- "I just showed them the data and they got it" — no evidence of tailoring
- Overcomplicating the explanation or hiding behind jargon
- No awareness that the audience had different priorities than the engineering team
- Blaming the audience for not understanding

Best Josh experiences:

- NASA decision matrices and trade-off documentation — literally built artifacts to communicate engineering rationale to external reviewers
 - NASA capstone as a whole — presenting to NASA stakeholders who care about mission success, not your CAD skills
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Question 10 — Solve Independently vs. Escalate

"Tell me about a situation where you had to decide whether to solve a problem on your own or escalate it to someone more experienced. What factors drove your decision?"

They're looking for:

- A clear framework for the decision — what factors tipped it one way or the other (time pressure, risk level, knowledge gap, impact of getting it wrong)
- Willingness to ask for help without ego
- Evidence they attempted to scope the problem before escalating, not just handing it off blindly

They don't want to see:

- "I always figure it out myself" — in field service, this gets you fired or gets equipment damaged
- "I always ask my manager" — no independence
- No reasoning behind the decision, just describing what happened
- Escalating without bringing any diagnostic information to the senior person

Best Josh experiences:

- Robotics lab research assistant — junior role working under a PI, natural dynamic of knowing what's in your lane vs. when to loop in the lead researcher
- NASA capstone with multiple subsystems — had to coordinate across teams, likely hit points where a decision was above his scope